

5. (a) A teacher showed her class how copper can be extracted from copper(II) oxide using charcoal.



In her experiment, the teacher heated 1.59 g of copper(II) oxide and 0.12 g of charcoal in a boiling tube. She continued to heat the mixture strongly until it had been glowing for 5 minutes.

She recorded the mass of the boiling tube and its contents before heating and then again after heating, once the carbon dioxide produced had been released from the tube.

Mass of boiling tube = 37.43 g

Mass of boiling tube and contents **before** heating = 39.14 g

Mass of boiling tube and contents **after** heating = 38.82 g

- (i) The teacher had expected the reaction to produce 1.27 g of copper from the masses of copper(II) oxide and charcoal used. Use her results to show that the mass produced suggests a 109 % yield. [2]

- (ii) Assuming the reactants had been heated at a high enough temperature for sufficient time, suggest **one** reason for the yield being larger than expected. [1]

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Examiner
only

(b) Iron is extracted from iron(III) oxide, Fe_2O_3 , inside the blast furnace using coke.

- (i) Write a balanced **symbol** equation for **one** of the reactions that show the reduction of iron(III) oxide inside the furnace. [2]

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- (ii) An iron ore contains 22% by mass of iron(III) oxide. Calculate the maximum mass of iron that could be obtained from 5×10^5 tonnes of this ore. Give your answer in **standard form**. [3]

Mass = tonnes

8



5. In 1985, a new allotrope of carbon was discovered and named buckminsterfullerene. This allotrope consists of sixty carbon atoms joined together to resemble a shape similar to that of a football (**Figure 1**), only ten septillion (10,000,000,000,000,000,000,000) times smaller.

This form of carbon was named after the architect Buckminster Fuller, famous for designing geodesic domes, as shown in **Figure 2**.

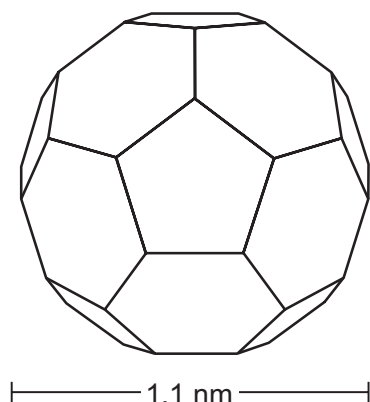


Figure 1



Figure 2

The structure of buckminsterfullerene is a truncated icosahedron, consisting of 20 hexagons and 12 pentagons that intersperse to form a spherical structure. Within the structure, each carbon atom is bonded to three other carbon atoms and no pentagon has a joining edge with another pentagon.

Other spherical allotropes of carbon, called fullerenes, have since been made. These include balls consisting of seventy, seventy-six and eighty-four carbon atoms. Together, they have become known as 'Buckyballs'.

Fullerenes have high melting points and boiling points. They also have a high density and a large surface area for their size.

Today, fullerenes are at the heart of nanotechnology – the study of atomic scale structures and devices. This provides many exciting new research possibilities for scientists including their potential uses in catalysts, lubricants and in nano-tubes for strengthening materials and as a way of delivering drugs into the body.

Nano-tubes are fullerenes that are used to reinforce graphite in tennis rackets because they are very strong. They are also used as semiconductors in electrical circuits.

The nano-tube's structure also allows it to be used as a container for transporting a drug in the body. A molecule of the drug can be placed inside the nano-tube cage. This keeps the drug 'wrapped up' until it reaches the site where it is needed. In this way, a dose that might be damaging to other parts of the body can be delivered safely to, for example, a tumour.



(a) Tick (✓) **all** the statements that correctly describe buckminsterfullerene.

[2]

its structure has 32 faces

☐

it has a relative molecular mass of 60

☐

it is an allotrope of carbon

☐

it has a giant ionic structure

☐

it is a hydrocarbon compound

☐

it is a smart material

☐

it is 1×10^{25} times smaller than a football

☐

(b) (i) Use the following formula to calculate the number of edges that a molecule of buckminsterfullerene has. [2]

$$\text{number of edges} = \frac{\text{total number of sides of all pentagons and hexagons}}{2}$$

Number of edges =



- (ii) Calculate, in m^3 , the approximate volume of a buckminsterfullerene molecule. Write your answer to **three** significant figures and in standard form. [3]

$$\text{volume} = \frac{4}{3} \pi r^3$$

$$\pi = 3.14$$

$$r = \text{radius}$$

$$1 \text{ nm} = 1 \times 10^{-9} \text{ m}$$

Volume = m^3

- (c) Give the **main** reason why the structure of fullerenes has resulted in there being an interest in developing their use as catalysts. [1]

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- (d) State why some people might oppose the use of fullerenes in drug delivery systems in the body. [1]

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- (e) One student said that Buckyballs should be good electrical conductors but her friend disagreed.

Use your knowledge of bonding and structure to give **one** reason that **each** student could use to support their argument. [2]

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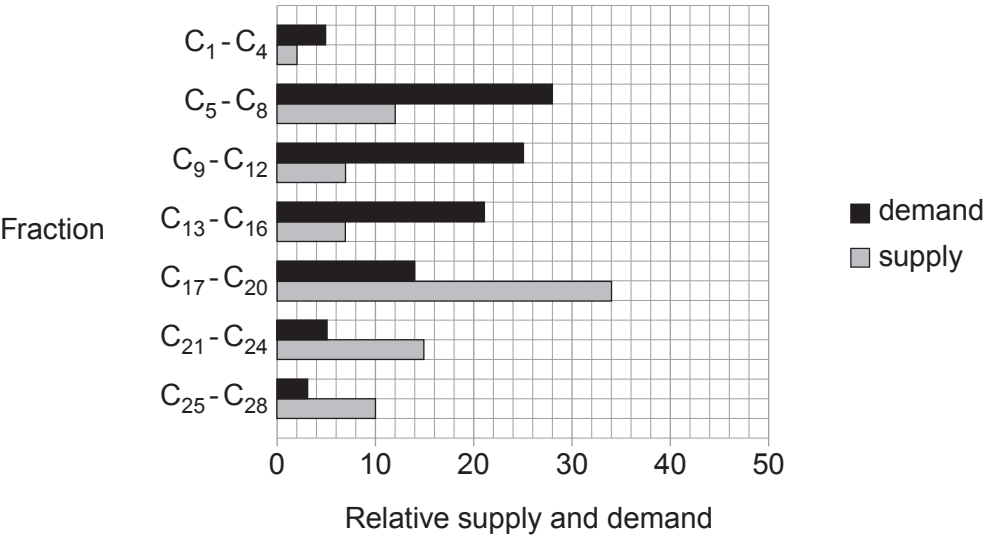
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5. (a) The bar chart shows the relative supply and demand for some fractions obtained from crude oil.



(i) Use the chart to describe how the **difference** between supply and demand of the fractions changes as the chain length increases. [2]

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- (ii) The process of cracking is used to overcome the problem of insufficient supply of some fractions.

I. State what is meant by *cracking* and give the conditions needed for the process. [2]

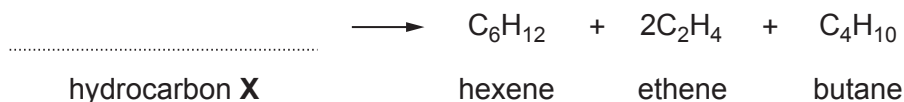
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II. When hydrocarbon **X** is cracked, it forms hexene, ethene and butane.

Give the molecular formula of hydrocarbon **X** to complete the equation. [1]



- (b) C_4H_{10} has two isomers.

(i) Give the meaning of the term *isomers*. [1]

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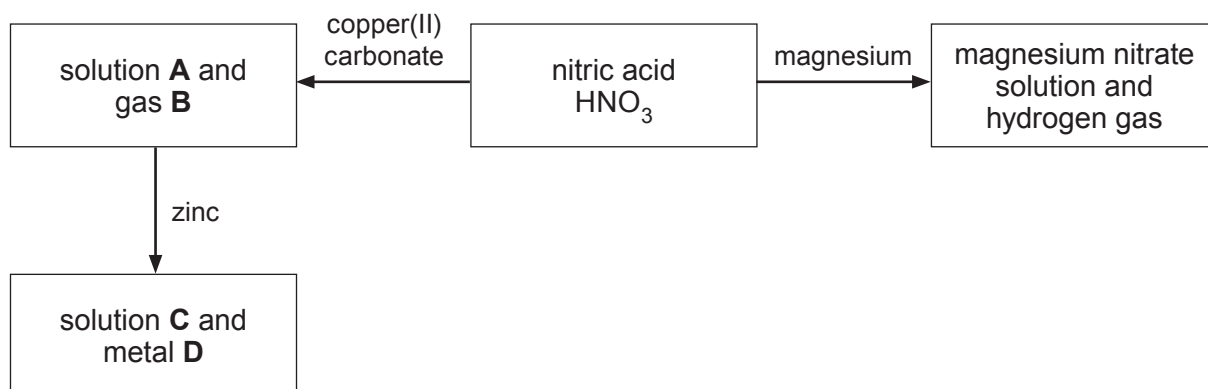
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(ii) Draw the structures of both isomers of C_4H_{10} . [2]



5. (a) The diagram shows a sequence of reactions from nitric acid.



- (i) Use the information in the diagram to name the following substances. [3]

solution **A**

gas **B**

solution **C**

metal **D**

- (ii) Write the balanced symbol equation for the reaction between magnesium and nitric acid. [3]

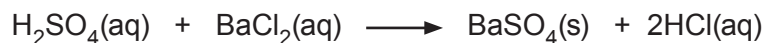
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- (iii) Name the type of reaction that takes place between solution **A** and zinc. [1]

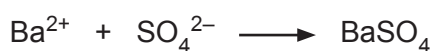
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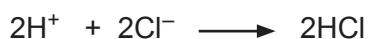
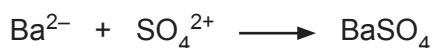


- (b) (i) The sulfate ions in sulfuric acid are identified using barium chloride solution. The equation for the reaction is shown below.



Tick (✓) the correct ionic equation for the formation of the precipitate. [1]


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- (ii) A student was asked to identify the ions present in barium chloride solution.

- I. Other than adding sulfate ions, state how the student would test for barium ions. Give the expected observation for a positive test. [1]

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- II. State how the student would test for chloride ions. Give the expected observation for a positive test. [1]

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